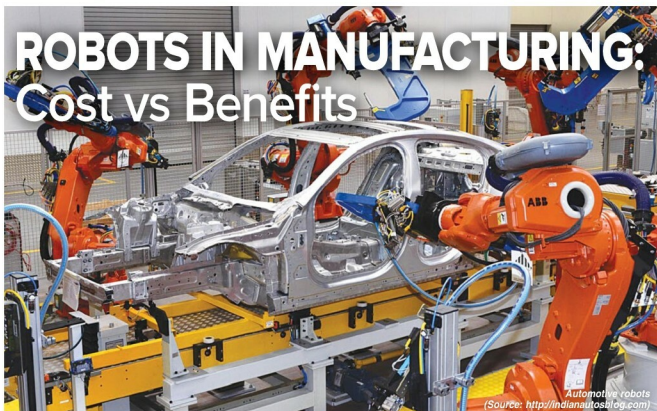




Automotive robots

(Source: <http://indianautosblog.com>)

ROBOTS IN MANUFACTURING: Cost vs Benefits

► Unlike human labour, robots can work three shifts a day, 365 days a year with very little down time

Automobile and electronics manufacturing in India is getting a facelift with heavy investments in robotic solutions. Government initiatives like 'Make in India' and 'Digital India' have given the trend a much-needed push.

Amit Maheshwari, Sastra Robotics, informs, "With a budget allocation of ₹ 1200 billion for the electronics manufacturing segment, and incentives to manufacturers for automation, robotics has received the required attention and push."

Purushothaman, EPR Labs, explains, "Homegrown brands like Flipkart are using artificial intelligence (AI) powered robots in their warehouses. With thousands of

new products being added every day, these AI empowered robots can segregate one brand of shampoo from another and easily categorise a new one based on its past learning from the environment and human interaction. AI-capable robots can cost as much as ₹ 4 million in the Indian market."

According to International Federation of Robotics, the investment in industrial robots in India will increase from 2126 units in 2014 to around 6000 units in 2019.

Ongoing Industrial Revolution 4.0, embracing automation, data exchange and manufacturing technologies, are all set to change the game and reinvent factories and production units.

When is the right time to use robotics?

According to Arjun Prakash, director, Efficacy Robots, there are three major reasons for manufacturing setups to invest in robotics:

1. Low efficiency due to too much labour dependence. On an average, robots replacing humans with the same cycle time tend to produce 25 per cent to 40 per cent more parts annually. This can be attributed to workers' fatigue, rest breaks and absence from work due to sick leaves or festival holidays. Most robots can work three shifts a day, 365 days a year with very little down time (which is mostly for annual greasing